The induced-module image is not B: $L \subseteq \ker M$ via a one-line sign argument

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Abstract

We retain notation: $\lambda \vdash n$ with $\ell = \ell(\lambda)$ rows; $H_q(S_n)$ acting on the Specht module V_λ in the Hoefsmit seminormal basis; $R'_j := (T_{j-1}+1)\cdots(T_1+1)$; $M := R'_{\ell+1}R'_{\ell+2}\cdots R'_n$; $B := M\,V_\lambda$; and the column-1-deletion shape $\lambda^{(\geq 2)} := (\lambda_1 - 1, \lambda_2 - 1, \ldots)$.

The May-14 PROVE.MD conjectured that B equals the image of the canonical Frobenius extension

$$\psi : \mathrm{Ind}_{H_q(S_\ell)\times H_q(S_{n-\ell})}^{H_q(S_n)}\big(V_{(1^\ell)} \boxtimes V_{\lambda^{(\geq 2)}}\big) \longrightarrow V_\lambda$$

of the LR-multiplicity-1 embedding. We refute this in the strongest possible sense:

Refutation. For irreducible V_λ at generic q, $\mathrm{im}(\psi) = V_\lambda$ entirely (since the image is an $H_q(S_n)$-submodule and $\psi \neq 0$). In every tested stable case, $\dim \mathrm{im}(\psi) = \dim V_\lambda \gg \dim B$ (e.g. 64 vs. 2 for $\lambda = (3,2,1,1,1)$ at $n = 8$).

Stronger structural finding. The LR-isotypic component

$$L \;:=\; V_{(1^\ell)} \boxtimes V_{\lambda^{(\geq 2)}} \;\hookrightarrow\; \mathrm{Res}_P V_\lambda$$

sits in the *kernel* of M, not in B: $L \subseteq \ker M$. Indeed, more generally:

Theorem 1 (Sign-kill lemma). *$E_1^-|_{V_\lambda} \subseteq \ker M$, where $E_1^- = \ker(T_1+1)$ is the (-1)-eigenspace of T_1.*

The proof is one line (each R'_j ends with (T_1+1) on the right; hence R'_n kills E_1^- and so does M).

This is a genuine *antipode* relationship: the subspace one would naively map to B via the leftmost-factor projector M is precisely the subspace M annihilates. Computationally we observe further $L \cap B = 0$ in every tested case, confirming that L and B are disjoint dim-$f^{\lambda^{(\geq 2)}}$ subspaces of V_λ — related by their dimension match (an LR coincidence) but otherwise unrelated. The induced-module route (MAY-13 resolution (c)) is therefore closed off; *any* representation-theoretic identification of B with $V_{\lambda^{(\geq 2)}}$ must use machinery other than the canonical Frobenius extension.

1 Setup

We retain the conventions of the May-13 multi-corner paper and the May-14 JM/non-stable note. Briefly:

- $H_q(S_n)$ has generators $T_1, \ldots, T_{n-1}$ subject to $(T_i - q)(T_i + 1) = 0$ and the braid relations.
- V_λ is the Specht module in the Hoefsmit seminormal basis $\{v_T : T \in \mathrm{SYT}(\lambda)\}$.
- For each $j \in \{1, \ldots, n-1\}$, $E_j^\pm \subseteq V_\lambda$ is the $(\pm)$-eigenspace of T_j (with eigenvalues q and -1 respectively).

- $R'_j := (T_{j-1}+1)(T_{j-2}+1)\cdots(T_1+1)$ for $j \geq 2$, and $R'_1 := 1$.
- For λ with $\ell = \ell(\lambda)$ rows, $M := R'_{\ell+1}R'_{\ell+2}\cdots R'_n$, and $B := \mathrm{im}(M) = M \cdot V_\lambda$.
- $\lambda^{(\geq 2)} := (\lambda_1 - 1, \lambda_2 - 1, \ldots, \lambda_\ell - 1)$ (drop zero parts), the column-1-deletion shape.
- $P := H_q(S_\ell) \times H_q(S_{n-\ell})$ is the standard parabolic (acting on positions $\{1, \ldots, \ell\}$ and $\{\ell + 1, \ldots, n\}$).

In the stable regime (May-13 closed-form theorem), $\dim B = f^{\lambda^{(\geq 2)}}$.

2 Refutation: the image of ψ is all of V_λ

The May-14 Prove.md proposed:

Conjecture 2 (May-14 Prove.md target)**.** *In the stable regime, the canonical $H_q(S_n)$-equivariant morphism*

$$\psi : \mathrm{Ind}_P^{H_q(S_n)}\big(V_{(1^\ell)} \boxtimes V_{\lambda^{(\geq 2)}}\big) \longrightarrow V_\lambda$$

has image equal to B.

2.1 Why ψ exists uniquely up to scalar

By Frobenius reciprocity for finite-dimensional Hopf-like algebras (in particular for $H_q(S_n)$, which is Frobenius and semisimple at generic q),

$$\mathrm{Hom}_{H_q(S_n)}\big(\mathrm{Ind}_P^{H_q(S_n)}(V_{(1^\ell)} \boxtimes V_{\lambda^{(\geq 2)}}),\ V_\lambda\big) \ \cong\ \mathrm{Hom}_P\big(V_{(1^\ell)} \boxtimes V_{\lambda^{(\geq 2)}},\ \mathrm{Res}_P V_\lambda\big).$$

The right-hand side has dimension equal to the multiplicity of $V_{(1^\ell)} \boxtimes V_{\lambda^{(\geq 2)}}$ in $\mathrm{Res}_P V_\lambda$, which by Littlewood–Richardson equals the LR coefficient $c^{\lambda}_{(1^\ell),\,\lambda^{(\geq 2)}}$. This coefficient is 1 (the skew shape $\lambda/(1^\ell)$ is a vertical strip, with the unique LR filling assigning content $\lambda^{(\geq 2)}$). Hence ψ exists and is unique up to scalar.

2.2 The image of ψ

Theorem 3 (Refutation)**.** *At generic q, $\mathrm{im}(\psi) = V_\lambda$. In particular, for every λ with $\dim V_\lambda > \dim B$ (which holds whenever λ is not an unusually small shape), $\mathrm{im}(\psi) \neq B$, refuting Conjecture 2.*

Proof. $\mathrm{im}(\psi) \subseteq V_\lambda$ is an $H_q(S_n)$-submodule. At generic q, $H_q(S_n)$ is semisimple and V_λ is irreducible, so the only submodules of V_λ are 0 and V_λ. Since $\psi \neq 0$ (LR multiplicity 1), $\mathrm{im}(\psi)$ is a nonzero submodule of V_λ, hence equal to V_λ. □ □

Remark 4 (Quantitative gap)**.** For $\lambda = (3, 2, 1, 1, 1)$ at $n = 8$ (a stable shape), $\dim V_\lambda = 64$ and $\dim B = 2$. So $\dim \mathrm{im}(\psi) = 64 \neq 2 = \dim B$. Similarly for every larger stable case in the May-13 family table.

The refutation is purely structural: the image of an equivariant map out of an induced module is always an honest submodule of V_λ, which by irreducibility forces it to be either zero or all of V_λ. The dimension match $\dim B = \dim L = f^{\lambda^{(\geq 2)}}$ is real but does not lift to an iso of ψ-images.

3 The sign-kill lemma

Theorem 5 (Sign-kill). *For any $\lambda \vdash n$ with $\ell = \ell(\lambda)$,*

$$E_1^-|_{V_\lambda} \subseteq \ker M.$$

That is, every vector $v \in V_\lambda$ with $T_1 v = -v$ satisfies $Mv = 0$.

Proof. The leftmost factor of R'_j in its standard expansion is $(T_{j-1}+1)$; the rightmost factor is (T_1+1). Hence for $j \geq 2$,

$$R'_j \;=\; X_j \cdot (T_1+1), \qquad X_j := (T_{j-1}+1)(T_{j-2}+1)\cdots(T_2+1) \;\in\; H_q(S_n).$$

For $v \in E_1^-$ (so $T_1 v = -v$), $(T_1+1)v = 0$, and therefore $R'_j v = X_j \cdot 0 = 0$ for every $j \geq 2$. In particular, $R'_n v = 0$, so

$$Mv \;=\; R'_{\ell+1}R'_{\ell+2}\cdots R'_{n-1}\cdot R'_n v \;=\; R'_{\ell+1}\cdots R'_{n-1}\cdot 0 \;=\; 0. \quad \square$$

$\square$

3.1 Corollaries

Corollary 6 ($L \subseteq \ker M$). *The LR-isotypic component $L := V_{(1^\ell)} \boxtimes V_{\lambda(\geq 2)} \subseteq \mathrm{Res}_P V_\lambda$ satisfies $L \subseteq \ker M$.*

Proof. $V_{(1^\ell)}$ is the sign representation of $H_q(S_\ell)$, on which T_i acts as -1 for every $i \in \{1,\ldots,\ell-1\}$. Hence the LR-isotypic component, viewed inside V_λ, satisfies $T_i v = -v$ for all $i \in \{1,\ldots,\ell-1\}$ and $v \in L$. In particular, $T_1 v = -v$, so $L \subseteq E_1^-$. By Theorem 5, $L \subseteq \ker M$. $\square$ $\square$

Corollary 7 (Combinatorial description of L in Hoefsmit basis). *In the Hoefsmit seminormal basis,*

$$L \;=\; \mathrm{span}\big\{v_T : T \in \mathrm{SYT}(\lambda),\ T(i,1) = i \textit{ for } i = 1,\ldots,\ell\big\}.$$

That is, L is the span of v_T for SYTs T whose column 1 is filled by $1,2,\ldots,\ell$ in order; equivalently, $T|_{[\ell]}$ is the unique SYT of shape (1^ℓ). In particular, $\dim L = f^{\lambda/(1^\ell)} = f^{\lambda(\geq 2)}$.

Proof. By Hoefsmit, the parabolic isotypic $S_\mu := \mathrm{span}\{v_T : T|_{[\ell]}$ has shape $\mu\}$ of $V_\lambda\!\downarrow_P$ is exactly $V_\mu \boxtimes V^{\lambda/\mu}$, where $V^{\lambda/\mu}$ is the skew Specht module. For $\mu = (1^\ell)$, $V^{\lambda/(1^\ell)} \cong V_{\lambda(\geq 2)}$ as $H_q(S_{n-\ell})$-modules (by the LR rule, the skew shape $\lambda/(1^\ell)$ has $V_{\lambda(\geq 2)}$ as its unique decomposition summand, with multiplicity 1). Hence $S_{(1^\ell)} \cong V_{(1^\ell)} \boxtimes V_{\lambda(\geq 2)} = L$ as P-modules, and the dimension count is the SYT count of the skew shape $\lambda/(1^\ell)$, which by translation equals $f^{\lambda(\geq 2)}$. $\square$ $\square$

3.2 L and B are disjoint in every tested case

Observation 8. In every stable case computed (Section 6),

$$L \cap B \;=\; 0.$$

That is, L and B are two distinct dim-$f^{\lambda(\geq 2)}$ subspaces of V_λ (with $L \subseteq \ker M$ by Corollary 6, and $B = \mathrm{im}(M)$ by definition), and they meet only at zero. The structural reason for this disjointness is presently unproven; the most natural attempt ($V_\lambda = \ker M \oplus \mathrm{im}(M)$, which would force $L \cap B \subseteq \ker M \cap \mathrm{im}(M) = 0$) fails in general because the Fitting decomposition of M is non-trivial: in some stable cases $M^2 = 0$ (e.g. $\lambda = (3,2,1,1,1)$, where $B \subseteq \ker M$ as well), and in others $\mathrm{rk}(M^2) = \mathrm{rk}(M)$ but M is not idempotent up to scalar. The disjointness $L \cap B = 0$ holds in both regimes.

λ	killed μ's	contributing μ's
$(3,2,1)$	(1^3)	$(2,1),\ (3)$
$(3,2,1,1)$	$(1^4),\ (2,1,1)$	$(2,2),\ (3,1)$
$(3,2,1,1,1)$	$(1^5),\ (2,1^3)$	$(2,2,1),\ (3,1,1),\ (3,2)$
$(3,3,1,1)$	$(1^4),\ (2,1,1)$	$(2,2),\ (3,1)$
$(4,2,1^4)$	$(1^6),\ (2,1^4),\ (2,2,1,1),\ (3,1,1,1)$	$(3,2,1),\ (4,1,1),\ (4,2)$
$(4,3,1^3)$	$(1^5),\ (2,1^3)$	$(2,2,1),\ (3,1,1),\ (3,2),\ (4,1)$

Table 1: The parabolic decomposition $V_\lambda{\downarrow_P}= \bigoplus_\mu S_\mu$: which μ have $M(S_\mu) = 0$ (*killed*) and which contribute nontrivially to $B = \sum_\mu M(S_\mu)$ (*contributing*). Computed mod $P = 100\,003$ at $q = 11$ via `probe_parabolic.py`.

4 The strict-superset structure of $\ker M$

Theorem 5 gives a sharp *lower bound*: $\ker M \supseteq E_1^-$. The kernel is in fact strictly larger; the gap is what makes $\dim B$ much smaller than $\dim E_1^+$.

Remark 9 ($E_i^- \not\subseteq \ker M$ for $i \geq 2$)**.** A computational check (Section 6) shows that $E_i^- \not\subseteq \ker M$ for any $i \geq 2$: in fact $\dim M(E_i^-) = \dim B$ for all $i \geq 2$ in the tested stable cases. So the sign-kill phenomenon is genuinely tied to position 1 and does not extend to other adjacent transpositions.

Remark 10 (The kernel via parabolic isotypics)**.** Decomposing $V_\lambda{\downarrow_P}= \bigoplus_{\mu\vdash\ell,\,\mu\subseteq\lambda} S_\mu$ with $S_\mu = V_\mu \boxtimes V^{\lambda/\mu}$, computational scan reveals:

$$M(S_\mu) \;=\; 0 \quad \text{for some structurally identifiable family of } \mu.$$

Specifically, $S_{(1^\ell)} = L \subseteq \ker M$ (this is Corollary 6), and the data shows that $S_{(2,1^{\ell-2})} \subseteq \ker M$ as well (in every tested case), along with several further μ's in larger shapes. The exact combinatorial criterion is open; the data is recorded in Table 1.

Conjecture 11 (Parabolic kill criterion, partial)**.** *There is a combinatorial criterion on μ (depending on λ) such that $S_\mu \subseteq \ker M$ for every μ failing the criterion. The criterion includes at minimum $\mu \in \{(1^\ell),\ (2,1^{\ell-2})\}$, since these always satisfy $T_1 v = -v$ (or have a related reduction to the same). A precise statement is open.*

5 Implications for the iso programme

1. **The induced-module route is closed.** The May-13 "resolution (c)" suggested that B might be the image of an equivariant morphism out of an induced module containing $V_{(1^\ell)} \boxtimes V_{\lambda(\geq 2)}$. Theorem 3 shows that the obvious such morphism (the canonical Frobenius extension ψ) has image equal to all of V_λ, not B. Combined with the May-13 refutation of standard parabolic embeddings and the May-14 refutation of seminormal-subset spans, this exhausts the natural representation-theoretic candidates for an iso $B \cong V_{\lambda(\geq 2)}$ via parabolic-style induction or restriction.

2. **The antipode picture.** The LR-isotypic L is the "natural input" to ψ (it is the embedded copy of $V_{(1^\ell)} \boxtimes V_{\lambda(\geq 2)}$ in $V_\lambda{\downarrow_P}$). Yet $L \subseteq \ker M$, so applying the leftmost-factor projector M

to L gives 0. Meanwhile $B = \mathrm{im}(M)$ is a different subspace of the same dimension. This is the strongest possible mismatch: L is exactly what M *annihilates*, while B is exactly what M *produces*. The dimension match is therefore a coincidence of LR-counting and $f^{\lambda^{(\geq 2)}}$-counting, with no direct map between L and B via the standard induced-module machinery.

3. **What might still work.** Three remaining routes are not obviously closed:

   (i) *The dual Frobenius map.* The companion morphism $\phi : V_\lambda \to \mathrm{Ind}_P^{H_q(S_n)}(V_{(1^\ell)} \boxtimes V_{\lambda^{(\geq 2)}})$ exists by the other Frobenius adjunction, with $\dim \mathrm{Hom}_G(V_\lambda, \mathrm{Ind}) = c^\lambda_{(1^\ell),\, \lambda^{(\geq 2)}} = 1$. By irreducibility, ϕ is injective. Whether ϕ^{-1} of some natural subspace of Ind equals B is open.

   (ii) *Cellular / KL bases.* The Hecke algebra $H_q(S_n)$ has a cellular basis (Kazhdan–Lusztig), and the seminormal Hoefsmit basis is one of several natural choices. The chains of the May-13 paper might lift to a different cellular description of B.

   (iii) *Subquotient construction.* B might admit a natural description as a subquotient of V_λ under a non-standard filtration, and the graded piece carrying $V_{\lambda^{(\geq 2)}}$ might be isomorphic to (a twist of) $V_{\lambda^{(\geq 2)}}$ as an $H_q(S_{n-\ell})$-module under *some* embedding.

6 Computational verification

All computations were performed mod $P = 100\,003$ at $q = 11$ using the seminormal-matrix infrastructure of the May-13 papers. Mod-P ranks coincide with $\mathbb{Q}(q)$-ranks at any non-special q; $q = 11$ is non-special for $|\lambda| \leq 11$.

6.1 Verification of $L \subseteq \ker M$ and $L \cap B = 0$

λ	n	$\dim V$	$\dim L$	$\dim B$	$\dim M(L)$	$\dim(L+B)$
$(3,2,1)$	6	16	2	2	0	4
$(3,2,1,1)$	7	35	2	2	0	4
$(3,2,1,1,1)$	8	64	2	2	0	4
$(3,3,1,1)$	8	56	2	2	0	4
$(4,2,1^4)$	10	350	3	3	0	6
$(4,3,1^3)$	10	525	5	5	0	10

Table 2: $\dim M(L) = 0$ in every case (confirming Corollary 6), and $\dim(L + B) = \dim L + \dim B$ in every case (confirming $L \cap B = 0$, Observation 8).

6.2 Verification of $E_1^- \subseteq \ker M$

6.3 Failure of E_i^- for $i \geq 2$

For the same shapes, $\dim M(E_i^-) = \dim B$ for every $i \in \{2, 3, \ldots, n-1\}$ — i.e., the operator M surjects from E_i^- onto B for $i \geq 2$, so E_i^- is *not* contained in $\ker M$.

λ	$\dim V$	$\dim B$	$\dim E_1^-$	$\dim M(E_1^-)$
$(3,2,1)$	16	2	8	0
$(3,2,1,1)$	35	2	20	0
$(3,2,1,1,1)$	64	2	40	0
$(3,3,1,1)$	56	2	30	0

Table 3: $M(E_1^-) = 0$ in every case (verifying Theorem 5 computationally).

6.4 Scripts

The computations were performed by:

- `/home/clio/projects/scratch/2026-05-14-induced-image/compare_L_B.py` — computes L, B, $M(L)$, $L \cap B$ for each test case.
- `/home/clio/projects/scratch/2026-05-14-induced-image/probe_parabolic.py` — decomposes $V_\lambda{\downarrow_P}$ into S_μ's and tests $\dim M(S_\mu)$ for each μ.
- `/home/clio/projects/scratch/2026-05-14-induced-image/probe_E_minus.py` — builds $E_i^- = \ker(T_i + 1)$ and tests $\dim M(E_i^-)$ for each i.
- `/home/clio/projects/scratch/2026-05-14-induced-image/probe_M_squared.py` — tests Fitting structure of M ($\operatorname{rk} M^2$ vs. $\operatorname{rk} M$).

7 Honest status

This note's main content is a refutation of the May-14 Prove.md target (Conjecture 2) at the strongest possible level: not only is $\operatorname{im}(\psi) \neq B$, but $\operatorname{im}(\psi) = V_\lambda$ entirely, and the would-be input L is precisely killed by M. Combined with the May-13 and May-14 refutations of standard parabolic embeddings and seminormal-subset spans, the natural "inductive / restricted Schur–Weyl" representation-theoretic identification of B with $V_{\lambda(\geq 2)}$ is now ruled out via every routine candidate.

The positive content is the one-line sign-kill lemma (Theorem 5), which makes the structural reason for $L \subseteq \ker M$ transparent and explains the parabolic kill of $S_{(1^\ell)}$ uniformly. The further parabolic kill data (Table 1) suggests a richer combinatorial criterion worth pursuing.

The remaining open route is the dual-Frobenius map $\phi : V_\lambda \to \operatorname{Ind}_P^G(V_{(1^\ell)} \boxtimes V_{\lambda(\geq 2)})$, which is injective by irreducibility; whether some natural pre-image under ϕ equals B is the leading remaining candidate, untested here.

A Why the standard symmetrizer-style identities don't help

For a Hecke algebra $H_q(S_k)$ with generators $T_1, \ldots, T_{k-1}$, the *trivial idempotent* (up to a scalar) admits the closed-form factorisation

$$h_k^+ \;=\; (T_{k-1}+1)(T_{k-2}T_{k-1} + T_{k-1} + T_{k-2} + 1)\cdots$$

or one of several Matsumoto-style products over the longest element. The element $R'_n = (T_{n-1} + 1)\cdots(T_1+1)$ used here is *not* the trivial idempotent; it is one factor of length $n-1$ (rather than $\binom{n}{2}$ as required), and its image is genuinely smaller than the trivial isotypic of $V_\lambda{\downarrow_{H_q(S_n)}}$.

Concretely, R'_n does not even square to a scalar multiple of itself in general; the May-14 `probe_M_squared.py` data shows $M^2 = 0$ for $\lambda = (3, 2, 1, 1, 1)$ (so $B \subseteq \ker M$ in that case) and $\operatorname{rk} M^2 = \operatorname{rk} M$ but $M^2 \neq cM$ for $\lambda = (3, 2, 1)$. The precise algebraic structure of M inside $H_q(S_n)$ is shape-dependent.